

SAI RAM PURIMETLA

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Education

New York University, Tandon School of Engineering

Master of Science in Computer Engineering | GPA: 3.80/4.0

May 2026

Brooklyn, NY

Experience

New York University, Tandon School of Engineering

Research Assistant

Brooklyn, NY

September 2025 – May 2026

- Developed an autonomous AI agent using Python and LangChain with structured self-correction for C/C++ kernel optimization, reducing manual development effort by 90%
- Engineered an automated LLM-integrated hardware synthesis pipeline in AMD Vitis HLS, eliminating manual data parsing and dynamically refining pragma directives to boost kernel efficiency by 40%
- Built validation and benchmarking infrastructure using Clang and TCL, accelerating synthesis error detection and extraction of PPA metrics across 500+ kernel evaluations

Amazon

Software Development Engineer Intern

Austin, TX

May 2025 – August 2025

- Engineered an event-driven incident triage pipeline using AWS SQS and SNS to trigger LangGraph workflows, autonomously retrieving contextual runbooks via a Retrieval-Augmented Generation (RAG) system and reducing on-call response latency by 50%
- Automated root-cause analysis by building LangGraph tool nodes that dynamically execute Model Context Protocol (MCP) workflows, optimizing AWS Athena log extractions to achieve a 40% reduction in query time and compute costs
- Synthesized actionable incident diagnostics using LangGraph reasoning agents to cross-reference extracted logs against runbook instructions, equipping on-call engineers with deep contextual insights to accelerate manual debugging
- Provisioned resilient Infrastructure-as-Code (IaC) using AWS CDK with TypeScript to deploy the LangGraph orchestration layer across auto-scaling AWS ECS Fargate clusters, guaranteeing 99.9% system availability under 2x projected traffic loads

Bosch Global Software Technologies

Software Engineer (Promoted from Associate)

Bangalore, India

July 2022 – August 2024

- Designed HLD/LLD and programmed a deterministic XML-to-C code generation pipeline using advanced Python and OOP, eliminating 95% of manual boilerplate and reducing post-integration defects
- Modernized legacy desktop applications and internal tools using PyQt5 and full-stack web frameworks (Django, Flask, HTML, CSS, SQL), enhancing system modularity and scalability
- Implemented unit and GUI automation frameworks using Pytest for BODAS IDE, increasing code coverage from 30% to 90% and strengthening release reliability
- Optimized high-volume 1TB XML workloads by implementing multi-threaded parsing and memory-mapped I/O, resolving memory leaks and driving a 70% increase in data throughput
- Established production CI/CD pipelines using Git, Jenkins, and Docker, remediating critical ETL pipeline failures and optimizing SQL query performance via advanced indexing and schema normalization, driving a 25% increase in sprint velocity and up to 60% faster query execution

Projects

Distributed Graph Service with High Availability | Java, Spring Boot, Redis, Cassandra, Kubernetes

December 2025

- Designed and implemented a scalable friend graph service supporting millions of users using Java and Spring Boot, providing low-latency relationship queries and updates
- Integrated Redis for caching frequently accessed graph edges and Cassandra for persistent storage, reducing database load by 70% and achieving sub-10ms p99 latency
- Deployed the service as microservices on Kubernetes (EKS) with horizontal pod autoscaling, circuit breakers, and readiness probes, maintaining 99.99% availability under peak traffic.

Enterprise RAG Knowledge Engine | Python, PyTorch, FastAPI, Vector DB, Redis, AWS ECS

May 2025

- Architected an end-to-end Retrieval-Augmented Generation (RAG) engine using dense embedding models and semantic chunking to vectorize enterprise data, achieving 98% retrieval accuracy for context-aware LLM generation
- Implemented Redis-based response caching to eliminate redundant LLM calls and reduce p95 latency by 40% for repeated queries
- Deployed a highly performant asynchronous FastAPI orchestration layer on AWS ECS with auto-scaling, optimizing vector similarity search and LLM inference bottlenecks to reduce end-to-end query latency to 400ms

Technical Skills

AI & Machine Learning: PyTorch, Transformers, LLMs, LangGraph, LangChain, RAG, Vector DBs, Agentic Workflows, CNNs, RNNs, GNNs

Languages: Python, Java, C/C++, TypeScript, SQL, HTML, CSS, TCL

Cloud & Tools: AWS (ECS, SQS, SNS, CDK, Athena), MCP, Docker, Kubernetes, Git, Jenkins, Pytest, FastAPI, Django, Flask, PyQt5, Redis, NoSQL (Cassandra, DynamoDB)

Core Concepts: Distributed Systems, Microservices, High Availability, System Design (HLD/LLD), Object-Oriented Programming, Multi-threading, Performance Optimization